

7. Determine the solution set for the inequality $\frac{x^3 + x^2 - 14x - 24}{-2x^2 + 8x + 10} \geq \frac{x+2}{x+1}$. Sketch and show all key features. Make sure to show all your work by hand and you may use desmos to help you.

$$\frac{x^3 + x^2 - 14x - 24}{-2(x-5)(x+1)} \geq \frac{x+2}{x+1}$$

$$\frac{x^3 + x^2 - 14x - 24 - (x+2)(x-5)(-2)}{(x+1)(-2)(x-5)} \geq 0$$

$$\frac{x^3 + x^2 - 14x - 24 - (-2x^2 + 6x + 20)}{-2(x+1)(x-5)} \geq 0$$

$$\frac{x^3 + 3x^2 - 20x - 44}{-2(x+1)(x-5)} \geq 0$$

$$-\frac{1}{2}x - \frac{7}{2} + \frac{13x-9}{-2x^2+8x+10} \geq 0$$

VA

$$x = -1$$

$$x = 5$$

oblique asymptote

$$x = -\frac{1}{2}x - \frac{7}{2}$$

$$x_{int} = 20$$

$$-2 \left| \begin{array}{ccc} 1 & 2 & -20 \\ 1 & 1 & -2 \end{array} \right| \begin{array}{c} -44 \\ 0 \end{array}$$

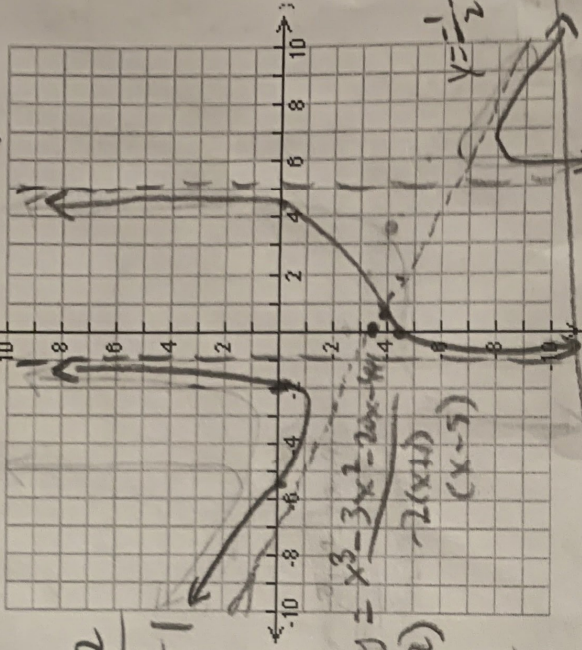
$$(x+2)(x^2+x-22)$$

$$x = -2$$

$$x = 1 \pm \sqrt{1-4(1)(-22)}$$

$$x = -5.217$$

$$x = -1$$



3.

$$\therefore \frac{x^3 + x^2 - 14x - 24}{-2x^2 + 8x + 10} \geq \frac{x+2}{x+1}$$

$$\text{where } x \in (-\infty, -5.217] \cup [4.217, 5)$$

$$-2 \leq x \leq -1 \quad 0 \leq x < 5$$

$$-\frac{1}{2}x - \frac{7}{2}$$

$$\frac{x^3 + 3x^2 - 20x - 44}{-2x^2 + 8x + 10} \geq 0$$

$$\begin{aligned} & \rightarrow x^3 - 4x^2 - 5x \\ & \rightarrow 7x^2 - 15x - 44 \\ & \rightarrow 7x^2 - 28x - 35 \end{aligned}$$

$$13x - 9$$

$$y = \frac{-44}{10} = -4.4$$

$$-5.217$$

$$4.217$$

1. Solve. Round answers to one decimal place.

a) $.01x^3 + .22x^2 + .2x = 3.3$

b) $2x^4 + x^3 - 26x^2 - 37x > 13$

$$x = 20.2$$

(3)

$$x \in (-\infty, -3.0) \cup$$

$$x = -5.0$$

$$(-0.91, -0.6) \cup (4.2, \infty)$$

2. Azar Corp was in business for 10 years from the year 2000 to 2010. Their profit equation can be modeled by the function $p(f) = 105x^3 - 1425x^2 + 4700x + 2500$. Determine the range of profits that Azar Corp experienced over the 10 years.

$$p \in [1578.849, 12000]$$

(2)

3. The specifications for a cardboard box state that the width is 5cm less than the length and the height is 1cm more than double the length. Find possible dimensions for a volume of 550cm^3 . (round final answer to two decimal places)

$$\text{length} = x$$

$$l = x = 8.55$$

possible dimensions

$$w = x - 5 = 3.55$$

$$h = 2x + 1 = 18.10$$

$$(x-5) \times (2x+1) = 550$$

(2)

$$8.55 \times 3.55 \times 18.1$$

$$F(t) = \frac{25t^2 + 125}{t^2 + 1}$$

4. The amount of food available on an island can be modeled by the function $F(t) = \frac{25t^2 + 125}{t^2 + 1}$, and the population on the same island can be modeled by the function $P(t) = 3t^2$ where t is the time in years, $0 \leq t \leq 10$.

- a) Describe what happens to the amount of food over the first 10 years.

It starts from 125 and then decreases at a rate that slows down until it gets to 25.99

(1)

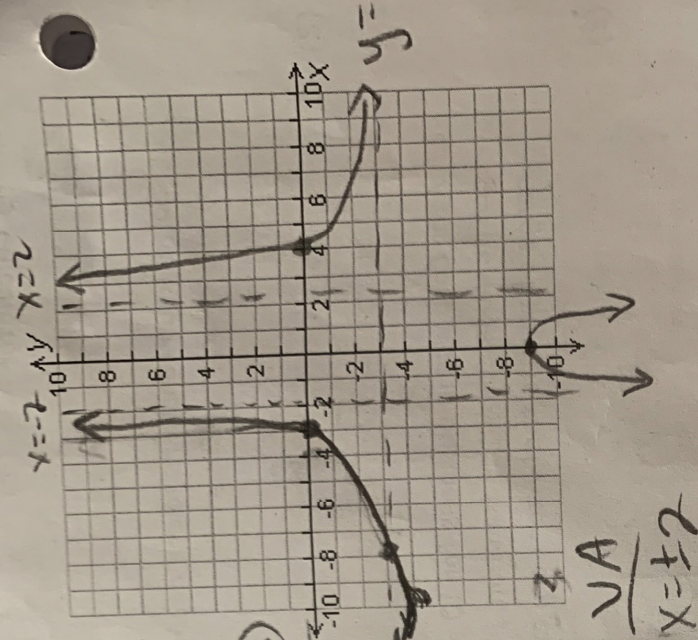
- b) When will the people on the island start running out of food?

They will when $p(t) > f(t)$ which happens when $t > 3.33$. So they will start

(1)

running out when $t = 3.33 \rightarrow 3.33$ years from now.

5. Graph the following function $f(x) = \frac{-3x^2 + 3x + 36}{x^2 - 4}$ on the grid provided showing all key features.



$$\begin{array}{l} x\text{-int} \\ x = -3, 4 \end{array}$$

$$\begin{array}{l} y\text{-int} \\ y = -9 \end{array}$$

intersection with HA

$$-3(x^2 - x - 12) = x^2 - 4$$

$$\begin{array}{l} -x = 8 \\ x = -8 \end{array}$$

$$\begin{array}{l} \frac{HA}{y} = -3 \end{array}$$

$$\begin{array}{l} \frac{VA}{x} = \pm 2 \end{array}$$

(6)

Domain

Range

$$x \in (-\infty, -2) \cup (-2, 2) \cup (2, \infty)$$

$$y \in (-\infty, 9] \cup$$

$$[-3.095, \infty)$$

6. In the previous question, determine the instantaneous rate of change at $x=5$. (2d.p)

$$\frac{f(5.001) - f(5)}{0.001}$$

(2)

$$= \frac{(-1.1436) - (-1.14286)}{0.001}$$

$$= -0.741232$$